14 (a) Simplify $\dfrac{x^2 - 16}{2x^2 - 5x - 12}$

×	x	$+4$
x	x^2	$+4x$
-4	$-4x$	-16

1 mark
Factorise the numerator

×	x	-4
$2x$	$2x^2$	$-8x$
$+3$	$+3x$	-12

1 mark
Factorise the denominator

$$\frac{\cancel{(x-4)}(x+4)}{\cancel{(x-4)}(2x+3)} = \frac{x+4}{2x+3}$$

1 mark
Cancel the common factor

Original expression: $x \neq 4$ and $x \neq -1.5$.

(3)

(b) Make v the subject of the formula $w = \dfrac{15(t - 2v)}{v}$

×	t	$-2v$
15	$15t$	$-30v$

$$wv = 15t - 30v$$
$$(+30v) \qquad (+30v)$$

1 mark
Multiply by v and expand the bracket

$$wv + 30v = 15t$$

×	w	30
v	wv	$30v$

$$v(w + 30) = 15t$$
$$(\div(w+30)) \qquad (\div(w+30))$$

1 mark
Collect the v terms and factor out v

$$v = \frac{15t}{w + 30}$$

1 mark
Make v the subject

$\dfrac{15t}{w + 30}$

(3)

(Total for Question 14 is 6 marks)

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